

ELITE MATHEMATICS

Pearson Edexcel International A Level

Mechanics 1

Adaptive Classified

Chapter 08 · Resolving Forces and Inclined Planes

With Answers

Questions with full worked solutions

- 24 pages
- Past papers up to June 2026

MECHANICS 1 | TOPIC 08

Resolving Forces, Inclined Planes

Your classified route

10

QUESTIONS

112

MARKS

2

FAMILIES

6

VARIANTS

DR ESLAM AHMED

CLASSIFIED BY IDEA

Read the diagram, choose the model, then use the working space to complete the mechanics.

TOPIC 08 | FAMILY 01

Motion on a rough inclined plane

WHAT THIS FAMILY ASKS

Resolve forces on a moving rough-plane particle and link $F = ma$ to later kinematics.

VARIANT ROUTE

V01 Single rough-plane particle

V02 Upward and downward stages

V03 Applied force/string removed

3 classified questions

39 total marks

Summer 2025 Q6 — complete original question

6.

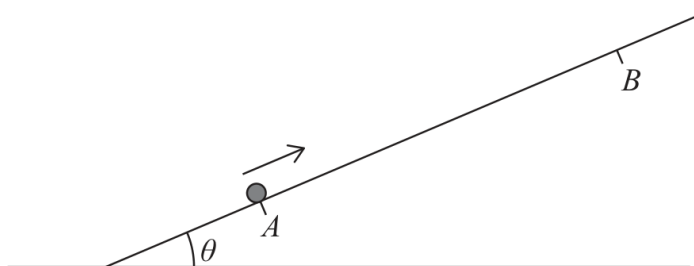


Figure 4

Figure 4 shows a rough plane inclined at an angle θ to the horizontal, where $\tan \theta = \frac{3}{4}$

The points A and B lie on a line of greatest slope of the plane, with B above A .

A package P of mass m is projected up the plane from A towards B .

The coefficient of friction between the plane and the package is $\frac{1}{4}$

The package P is modelled as a particle.

- (a) Show that the **deceleration** of P , as it moves from A to B , is $\frac{4}{5}g$ (6)

The package P comes to rest at B .

Given that P is projected from A with speed $U \text{ m s}^{-1}$ and that $AB = 1.5 \text{ m}$,

- (b) find the value of U . (2)

On reaching B , P is held at rest there by a force of magnitude X newtons acting up the plane in the direction AB .

Given that the mass of P is 2 kg ,

- (c) find the smallest possible value of X . (3)

Summer 2025 Q6 — detailed solution

Single rough-plane particle

ALL
11 marks**Part (a): Show deceleration = $\frac{4g}{5}$ - Find exact trig values**Given: $\tan \theta = \frac{3}{4}$ Using 3-4-5 right triangle: $\sin \theta = \frac{3}{5}$, $\cos \theta = \frac{4}{5}$ **Part (a): Show deceleration = $\frac{4g}{5}$ - Resolve perpendicular to plane** $\perp: \sum F = 0$

$$R = mg \cos \theta \quad R = \frac{4mg}{5}$$

Part (a): Show deceleration = $\frac{4g}{5}$ - Calculate friction force

Package moving UP, so friction acts DOWN the plane:

$$F = \mu R = \frac{1}{4} \times \frac{4mg}{5} = \frac{mg}{5}$$

Part (a): Show deceleration = $\frac{4g}{5}$ - Resolve parallel to plane

Taking up the plane as positive:

$$\parallel: \sum F = ma$$

$$-mg \sin \theta - F = ma \quad -mg \times \frac{3}{5} - \frac{mg}{5} = ma \quad -\frac{3mg}{5} - \frac{mg}{5} = ma \quad -\frac{4mg}{5} = ma \quad a = -\frac{4g}{5}$$

$$\text{Deceleration} = \left| -\frac{4g}{5} \right| = \frac{4g}{5}$$

Part (b): Find initial speed U - Identify SUVAT variables

Package projected up plane, comes to rest at B:

 $u = U \text{ m s}^{-1}$ (unknown) ; $v = 0 \text{ m s}^{-1}$ (comes to rest) ; $a = -\frac{4g}{5} \text{ m s}^{-2}$ (from part a) ; $s = 1.5 \text{ m}$ **Part (b): Find initial speed U - Apply $v^2 = u^2 + 2as$**

$$v^2 = u^2 + 2as$$

$$0^2 = U^2 + 2 \left(-\frac{4g}{5} \right) (1.5) \quad 0 = U^2 - 2 \times \frac{4 \times 9.8}{5} \times 1.5 \quad 0 = U^2 - \frac{8 \times 9.8 \times 1.5}{5} \quad 0 = U^2 - 23.52 \quad U^2 = 23.52$$

$$U = 4.85 \text{ m s}^{-1}$$

$$\text{Or using exact form: } U = \sqrt{2 \times \frac{4g}{5} \times 1.5} = \sqrt{\frac{12g}{5}}$$

Part (c): Find smallest value of X - Analyze forces when package at rest at B

Package held at rest on slope by force X up the plane.

Without X, package would slide down (component down > friction up).

For MINIMUM X, friction acts UP the plane (at limiting friction).

Part (c): Find smallest value of X - Resolve perpendicular to plane

Mass = 2 kg, so weight = 2g N

$$R = 2g \cos \theta = 2g \times \frac{4}{5} = \frac{8g}{5} \text{ N}$$

Part (c): Find smallest value of X - Calculate friction

Friction acts UP the plane (limiting):

$$F = \mu R = \frac{1}{4} \times \frac{8g}{5} = \frac{2g}{5} \text{ N}$$

Part (c): Find smallest value of X - Resolve parallel to plane

For equilibrium (package at rest):

$$\parallel: \sum F = 0$$

Taking up plane as positive:

$$X + F - 2g \sin \theta = 0 \quad X + \frac{2g}{5} - 2g \times \frac{3}{5} = 0 \quad X + \frac{2g}{5} - \frac{6g}{5} = 0 \quad X = \frac{6g}{5} - \frac{2g}{5} \quad X = \frac{4g}{5} \quad X = \frac{4 \times 9.8}{5} \quad X = 7.84 \text{ N}$$

Final answer (a) deceleration = $\frac{4g}{5}$, as required. (b) $U = 4.85 \text{ m s}^{-1}$ (about 4.8 m s^{-1}). (c) $X = 7.84 \text{ N}$ (about 7.8 N).

DR ESLAM AHMED

Summer 2021 Q6 — complete original question

6. A fixed rough plane is inclined at an angle θ to the horizontal, where $\tan \theta = \frac{5}{12}$

A particle of mass 6 kg is projected with speed 5 m s^{-1} from a point A on the plane, up a line of greatest slope of the plane.

The coefficient of friction between the particle and the plane is $\frac{1}{4}$

- (a) Find the magnitude of the frictional force acting on the particle as it moves up the plane.

(3)

The particle comes to instantaneous rest at the point B .

- (b) Find the distance AB .

(5)

The particle now slides down the plane from B . At the instant when the particle passes through the point C on the plane, the speed of the particle is again 5 m s^{-1}

- (c) Find the distance BC .

(5)

Summer 2021 Q6 — detailed solution

Upward and downward stages

ALL
13 marks**Use the trig ratios**

Since $\tan \theta = \frac{5}{12}$, $\sin \theta = \frac{5}{13}$ and $\cos \theta = \frac{12}{13}$.

Find the normal reaction

$$R = 6g \cos \theta = 6g \cdot \frac{12}{13} = \frac{72g}{13}.$$

Find friction

With $\mu = \frac{1}{4}$, $F = \frac{1}{4}R = \frac{18g}{13} = 13.6 \text{ N}$ approximately.

Find acceleration while moving up

While moving up, both friction and the component of weight act down the plane. Taking up the plane as positive: $6a = -6g \sin \theta - F$.

Simplify the upward acceleration

$$6a = -6g \cdot \frac{5}{13} - \frac{18g}{13} = -\frac{48g}{13}, \text{ so } a = -\frac{8g}{13}.$$

Find AB

Use $v^2 = u^2 + 2as$ with $u = 5$ and $v = 0$: $0 = 25 + 2 \left(-\frac{8g}{13} \right) AB$. Hence $AB = 2.07 \dots \text{ m}$.

Find acceleration while sliding down

When sliding down, friction acts up the plane. Taking down the plane as positive: $6a = 6g \sin \theta - F = \frac{30g}{13} - \frac{18g}{13} = \frac{12g}{13}$. Thus $a = \frac{2g}{13}$.

Find BC

Starting from rest at B and reaching speed 5 at C: $25 = 2 \left(\frac{2g}{13} \right) BC$. Hence $BC = 8.29 \dots \text{ m}$.

Final answer (a) $F = 13.6 \text{ N}$. (b) $AB = 2.07 \text{ m} \approx 2.1 \text{ m}$. (c) $BC = 8.29 \text{ m} \approx 8.3 \text{ m}$.

January 2023 Q8 — complete original question

8.

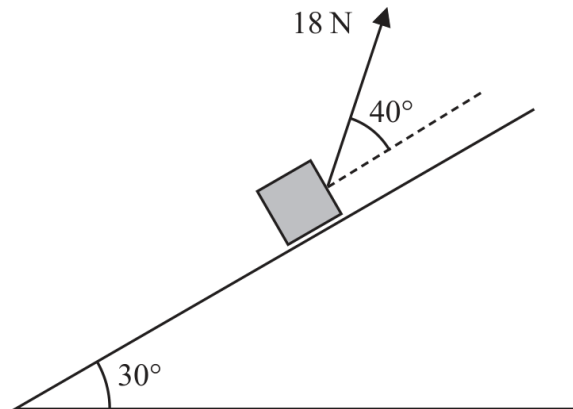


Figure 5

A parcel of mass 2 kg is pulled up a rough inclined plane by the action of a constant force.

The force has magnitude 18 N and acts at an angle of 40° to the plane.

The line of action of the force lies in a vertical plane containing a line of greatest slope of the inclined plane.

The plane is inclined at an angle of 30° to the horizontal, as shown in Figure 5.

The coefficient of friction between the plane and the parcel is 0.3

The parcel is modelled as a particle P

(a) Find the acceleration of P (8)

The points A and B lie on a line of greatest slope of the plane, where $AB = 5$ m and B is above A . Particle P passes through A with speed 2 m s^{-1} in the direction AB .

(b) Find the speed of P as it passes through B . (3)

The force of 18 N is removed at the instant P passes through B . As a result, P comes to rest at the point C .

(c) Determine whether P will remain at rest at C . You must show all stages of your working clearly. (4)

January 2023 Q8 — detailed solution

Applied force/string removed

ALL
15 marks**Resolve perpendicular to the plane**

The force of 18 N is angled away from the plane, so $R + 18 \sin 40^\circ = 2g \cos 30^\circ$. Hence $R = 2g \cos 30^\circ - 18 \sin 40^\circ$.

Use friction

The parcel moves up the plane, so friction acts down the plane. With $\mu = 0.3$, $F = 0.3R$.

Resolve parallel to the plane

Taking up the plane as positive: $18 \cos 40^\circ - F - 2g \sin 30^\circ = 2a$.

Substitute for R and F

$18 \cos 40^\circ - 0.3(2g \cos 30^\circ - 18 \sin 40^\circ) - 2g \sin 30^\circ = 2a$.

Find the acceleration

Using $g = 9.8$, this gives $a = 1.18 \dots \text{ m s}^{-2}$ up the plane.

Find the speed at B

From A to B, $u = 2$, $s = 5$ and $a = 1.18 \dots$. Use $v^2 = u^2 + 2as$: $v^2 = 2^2 + 2(1.18 \dots)(5)$.

State the speed

$v = 3.98 \dots \text{ m s}^{-1}$, about 4.0 m s^{-1} .

Check rest after the force is removed

At C, with the pulling force removed, $R = 2g \cos 30^\circ$. The maximum friction is $0.3R = 0.3(2g \cos 30^\circ) = 5.09 \dots \text{ N}$.

Compare with the downslope weight component

The component of weight down the plane is $2g \sin 30^\circ = 9.8 \text{ N}$. Since $9.8 > 5.09 \dots$, friction is not large enough to hold P at rest, so P will not remain at rest at C.

Final answer (a) $a = 1.18 \text{ m s}^{-2}$ up the plane. (b) $v = 3.98 \text{ m s}^{-1} \approx 4.0 \text{ m s}^{-1}$. (c) P will not remain at rest at C.

TOPIC 08 | FAMILY 02

Limiting equilibrium on a rough inclined plane

WHAT THIS FAMILY ASKS

Resolve a rough-plane particle in equilibrium and apply $F \leq \mu R$ or $F = \mu R$ at a boundary.

VARIANT ROUTE

V01 Impending motion up or down the plane

V02 Two opposite rough-plane limiting states

V03 Oblique/horizontal force changes a rough-plane reaction

7 classified questions

73 total marks

January 2024 Q8 — complete original question

8.

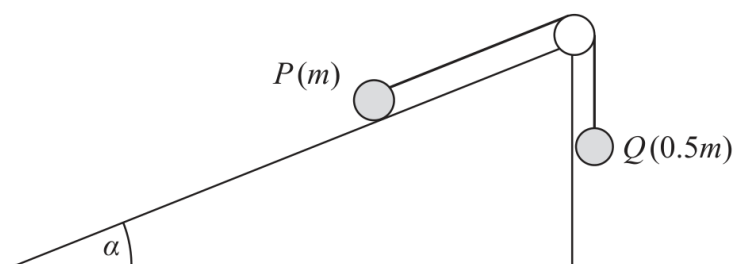


Figure 5

A fixed rough plane is inclined at an angle α to the horizontal, where $\tan \alpha = \frac{5}{12}$

A small smooth pulley is fixed at the top of the plane.

One end of a light inextensible string is attached to a particle P which is at rest on the plane. The string passes over the pulley and the other end of the string is attached to a particle Q which hangs vertically below the pulley, as shown in Figure 5.

Particle P has mass m and particle Q has mass $0.5m$

The string from P to the pulley lies along a line of greatest slope of the plane.

The coefficient of friction between P and the plane is μ .

The system is in **limiting equilibrium** with the string taut and P is on the point of slipping **up** the plane.

(a) Find the value of μ .

(8)

The string breaks and P begins to move down the plane.

When particle P has travelled a distance of 0.8 m down the plane, the speed of P is $V \text{ m s}^{-1}$

(b) Find the value of V .

(4)

January 2024 Q8 — detailed solution**ALL**
12 marks**Use the trig ratios**

Given $\tan \alpha = \frac{5}{12}$, use $\sin \alpha = \frac{5}{13}$ and $\cos \alpha = \frac{12}{13}$.

Find the tension

Q has mass $0.5m$ and the system is in equilibrium, so the tension is $T = 0.5mg$.

Resolve P parallel to the plane

P is on the point of slipping up the plane, so friction acts down the plane. Along the plane, $T = mg \sin \alpha + F$.

Find the friction force

$$F = T - mg \sin \alpha = \frac{1}{2}mg - \frac{5}{13}mg = \frac{3}{26}mg.$$

Find the normal reaction and μ

$$R = mg \cos \alpha = \frac{12}{13}mg. \text{ Since } F = \mu R, \mu = \frac{3mg/26}{12mg/13} = \frac{1}{8}.$$

Find acceleration after the string breaks

P moves down the plane, so friction acts up the plane. The resultant down the plane is $mg \sin \alpha - F = \frac{5}{13}mg - \frac{1}{8} \cdot \frac{12}{13}mg = \frac{7}{26}mg$.

Apply Newtons second law

As P has mass m , $a = \frac{7g}{26}$ down the plane.

Find V

From rest over 0.8 m, $V^2 = 2 \left(\frac{7g}{26} \right) (0.8)$. With $g = 9.8$, $V = 2.05 \text{ m s}^{-1}$ to 3 significant figures.

Final answer (a) $\mu = \frac{1}{8}$. (b) $V = 2.05 \text{ m s}^{-1}$ approximately.

January 2022 Q5 — complete original question

5.

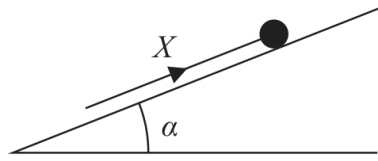


Figure 3

A particle of mass m rests in equilibrium on a fixed rough plane under the action of a force of magnitude X . The force acts up a line of greatest slope of the plane, as shown in Figure 3.

The plane is inclined at an angle α to the horizontal, where $\tan \alpha = \frac{3}{4}$

The coefficient of friction between the particle and the plane is μ .

- When $X = 2P$, the particle is on the point of sliding up the plane.
- When $X = P$, the particle is on the point of sliding down the plane.

Find the value of μ .

(8)

January 2022 Q5 — detailed solution

Two opposite rough-plane limiting states

ALL
8 marks**Use the trig ratios**

Since $\tan \alpha = \frac{3}{4}$, $\sin \alpha = \frac{3}{5}$ and $\cos \alpha = \frac{4}{5}$.

Resolve perpendicular to the plane

The force X acts along the plane, so the normal reaction is $R = mg \cos \alpha = \frac{4}{5}mg$.

Use friction

The limiting friction is $F = \mu R = \frac{4}{5}\mu mg$.

Use the sliding-up condition

When $X = 2P$, the particle is about to slide up, so friction acts down the plane. Hence $2P = mg \sin \alpha + F = \frac{3}{5}mg + \frac{4}{5}\mu mg$.

Use the sliding-down condition

When $X = P$, the particle is about to slide down, so friction acts up the plane. Hence $P + F = mg \sin \alpha$, so $P = \frac{3}{5}mg - \frac{4}{5}\mu mg$.

Eliminate P

Double the second equation: $2P = \frac{6}{5}mg - \frac{8}{5}\mu mg$. This equals the first expression for $2P$.

Solve for mu

$\frac{6}{5}mg - \frac{8}{5}\mu mg = \frac{3}{5}mg + \frac{4}{5}\mu mg$. Cancel mg and multiply by 5: $6 - 8\mu = 3 + 4\mu$. Hence $3 = 12\mu$ and $\mu = \frac{1}{4}$.

Final answer $\mu = \frac{1}{4}$.

October 2019 Q4 — complete original question

4.

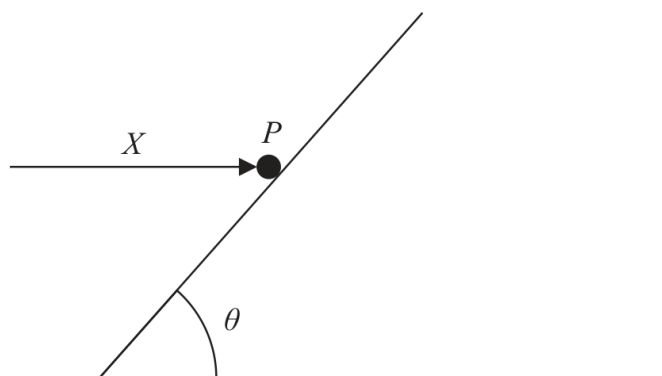


Figure 1

A particle, P , of mass km lies on a fixed rough plane. The plane is inclined to the horizontal at an acute angle θ . A horizontal force of magnitude X acts on P , as shown in Figure 1. The line of action of the force lies in the vertical plane which contains the line of greatest slope of the inclined plane that passes through P . The coefficient of friction between P and the inclined plane is μ .

When $X = mg$, the particle P is in equilibrium and on the point of sliding down the plane.

(a) Show that $\mu = \frac{k \tan \theta - 1}{k + \tan \theta}$ (10)

(b) Deduce that, when $k = 1$, θ must be greater than 45° (2)

October 2019 Q4 — detailed solution

Oblique/horizontal force changes a rough-plane reaction

ALL
12 marks**Set friction direction**

The particle is on the point of sliding down the plane, so friction acts up the plane. Also $X = mg$.

Resolve perpendicular to the plane

Both the weight kmg and the horizontal force mg press the particle into the plane. Therefore $R = kmg \cos \theta + mg \sin \theta$.

Resolve parallel to the plane

Down the plane is the component $kmg \sin \theta$. Up the plane are friction F and the component $mg \cos \theta$. Thus $F + mg \cos \theta = kmg \sin \theta$.

Write friction in terms of theta

So $F = kmg \sin \theta - mg \cos \theta$.

Use limiting friction

$$\mu = \frac{F}{R} = \frac{kmg \sin \theta - mg \cos \theta}{kmg \cos \theta + mg \sin \theta}$$

Simplify the expression

Cancel mg and divide numerator and denominator by $\cos \theta$: $\mu = \frac{k \tan \theta - 1}{k + \tan \theta}$, as required.

Deduce the condition when k is 1

When $k = 1$, $\mu = \frac{\tan \theta - 1}{1 + \tan \theta}$. Since $\mu > 0$ and the denominator is positive, $\tan \theta - 1 > 0$.

Finish the inequality

Therefore $\tan \theta > 1$, so because θ is acute, $\theta > 45^\circ$.

Final answer (a) $\mu = \frac{k \tan \theta - 1}{k + \tan \theta}$. (b) $\theta > 45^\circ$.

October 2022 Q3 — complete original question

3.

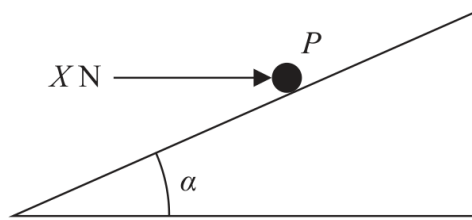


Figure 2

A rough plane is inclined to the horizontal at an angle α , where $\tan \alpha = \frac{3}{4}$

A particle P of mass 2 kg is held in equilibrium on the plane by a horizontal force of magnitude X newtons, as shown in Figure 2. The force acts in a vertical plane which contains a line of greatest slope of the inclined plane.

(a) Show that when $X = 14.7$ there is no frictional force acting on P (3)

The coefficient of friction between P and the plane is 0.5

(b) Find the smallest possible value of X . (8)

October 2022 Q3 — detailed solution

Oblique/horizontal force changes a rough-plane reaction

ALL
11 marks**Use the trig ratios**

Since $\tan \alpha = \frac{3}{4}$, $\sin \alpha = \frac{3}{5}$ and $\cos \alpha = \frac{4}{5}$.

Check the no-friction case

If there is no friction, the component of the horizontal force up the plane balances the component of weight down the plane: $X \cos \alpha = 2g \sin \alpha$.

Substitute $X = 14.7$

$14.7 \cdot \frac{4}{5} = 11.76$, and $2(9.8) \cdot \frac{3}{5} = 11.76$. The components balance exactly, so the frictional force is zero.

Set the direction for smallest X

For the smallest possible X , the particle is about to slide down the plane, so friction acts up the plane.

Resolve perpendicular to the plane

The horizontal force increases the normal reaction, so $R = 2g \cos \alpha + X \sin \alpha$.

Use limiting friction

With $\mu = 0.5$, the limiting friction is $F = 0.5R = 0.5(2g \cos \alpha + X \sin \alpha)$.

Resolve along the plane

At the limiting case, $X \cos \alpha + F = 2g \sin \alpha$.

Solve for X

Substitute the exact trig values: $\frac{4}{5}X + 0.5 \left(2g \cdot \frac{4}{5} + X \cdot \frac{3}{5} \right) = 2g \cdot \frac{3}{5}$. This gives $\frac{11}{10}X + \frac{4g}{5} = \frac{6g}{5}$, so

$$X = \frac{4g}{11} = 3.56 \dots \text{ N.}$$

Final answer (a) When $X = 14.7$, $F = 0$. (b) $X_{\min} = \frac{4g}{11} = 3.56 \dots \text{ N.}$

October 2024 Q6 — complete original question

6.

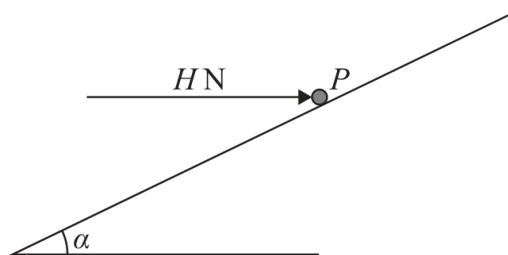


Figure 4

A particle P of mass 5 kg lies on the surface of a rough plane.

The plane is inclined at an angle α to the horizontal, where $\tan \alpha = \frac{3}{4}$

The particle is held in equilibrium by a horizontal force of magnitude H newtons, as shown in Figure 4.

The horizontal force acts in a vertical plane containing a line of greatest slope of the inclined plane.

The coefficient of friction between the particle and the plane is $\frac{1}{4}$

(a) Find the smallest possible value of H .

(6)

The horizontal force is now removed, and P starts to slide down the slope.

In the first T seconds after P is released from rest, P slides 1.5 m down the slope.

(b) Find the value of T .

(6)

October 2024 Q6 — detailed solution

Oblique/horizontal force changes a rough-plane reaction

ALL
12 marks**Use the triangle for alpha**

From $\tan \alpha = \frac{3}{4}$, use $\sin \alpha = \frac{3}{5}$ and $\cos \alpha = \frac{4}{5}$.

Resolve perpendicular to the plane

The horizontal force has a component into the plane, so $R = 5g \cos \alpha + H \sin \alpha = 4g + \frac{3}{5}H$.

Resolve parallel for the smallest H

For the smallest H , the particle is on the point of sliding down, so friction acts up the plane. Parallel to the plane, $H \cos \alpha + F = 5g \sin \alpha$.

Use friction to find H

With $F = \frac{1}{4}R$, the parallel equation becomes $\frac{4}{5}H + \frac{1}{4}\left(4g + \frac{3}{5}H\right) = 3g$. Hence $0.95H + g = 3g$, so $H = \frac{40g}{19} = 20.6 \text{ N}$.

Find the acceleration after H is removed

When H is removed, $R = 5g \cos \alpha = 4g$ and friction is $\frac{1}{4}R = g$. The component of weight down the plane is $5g \sin \alpha = 3g$, so the resultant down the plane is $2g$.

Apply Newtons second law

$2g = 5a$, so $a = \frac{2g}{5} = 3.92 \text{ m s}^{-2}$ down the plane.

Find T from the displacement

From rest, the particle moves 1.5 m with acceleration $2g/5$. Using $s = \frac{1}{2}at^2$: $1.5 = \frac{1}{2}\left(\frac{2g}{5}\right)T^2 = \frac{g}{5}T^2$. Thus $T = \sqrt{\frac{7.5}{9.8}} = 0.875 \text{ s}$.

Final answer (a) $H = \frac{40g}{19} \text{ N} \approx 20.6 \text{ N}$. (b) $T = 0.875 \text{ s}$.

January 2025 Q6 — complete original question

6.

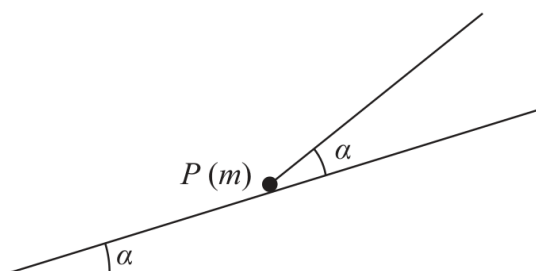


Figure 3

A particle P of mass m is held in equilibrium on a fixed rough inclined plane by a light inextensible string.

The plane is inclined at an angle α to the horizontal, where $\alpha < 45^\circ$.
The string is inclined to the plane at angle α , as shown in Figure 3.

The string lies in a vertical plane that contains a line of greatest slope of the inclined plane.

When the tension in the string is $0.75mg$, P is on the point of moving up the plane.

- (a) Find an expression for the magnitude of the frictional force acting on P , giving your answer in terms of m , g and α .

(3)

The coefficient of friction between P and the plane is $\frac{1}{2}$.

- (b) Show that

$$\tan \alpha = \frac{2}{5}$$

(6)

The string breaks.

- (c) Determine whether P remains at rest. You must justify your reasoning.

(3)

January 2025 Q6 — detailed solution

Oblique/horizontal force changes a rough-plane reaction

ALL
12 marks**Resolve parallel to the plane**

P is on the point of moving up the plane, so friction acts down the plane. The tension has component $0.75mg \cos \alpha$ up the plane, and the weight component is $mg \sin \alpha$ down the plane.

Find the frictional force

For equilibrium parallel to the plane, $0.75mg \cos \alpha = mg \sin \alpha + F$. Hence $F = 0.75mg \cos \alpha - mg \sin \alpha$.

Resolve perpendicular to the plane

The string has a component away from the plane, so $R = mg \cos \alpha - 0.75mg \sin \alpha$.

Use the coefficient of friction

Given $\mu = \frac{1}{2}$, limiting friction gives $F = \frac{1}{2}R$. Therefore $0.75mg \cos \alpha - mg \sin \alpha = \frac{1}{2}(mg \cos \alpha - 0.75mg \sin \alpha)$.

Show the tangent result

Divide by mg and collect terms: $0.75 \cos \alpha - \sin \alpha = 0.5 \cos \alpha - 0.375 \sin \alpha$. Thus $0.25 \cos \alpha = 0.625 \sin \alpha$, so $\tan \alpha = \frac{0.25}{0.625} = \frac{2}{5}$.

Check what happens when the string breaks

Without the string, $R = mg \cos \alpha$ and the maximum friction is $\frac{1}{2}mg \cos \alpha$. The component of weight down the plane is $mg \sin \alpha$.

Compare friction with the downslope component

Since $\tan \alpha = \frac{2}{5} < \frac{1}{2}$, we have $mg \sin \alpha < \frac{1}{2}mg \cos \alpha$. The available friction is large enough to hold P, so P remains at rest.

Final answer (a) $F = 0.75mg \cos \alpha - mg \sin \alpha$. (b) $\tan \alpha = \frac{2}{5}$. (c) P remains at rest.

Summer 2026 Q5 — complete original question

5

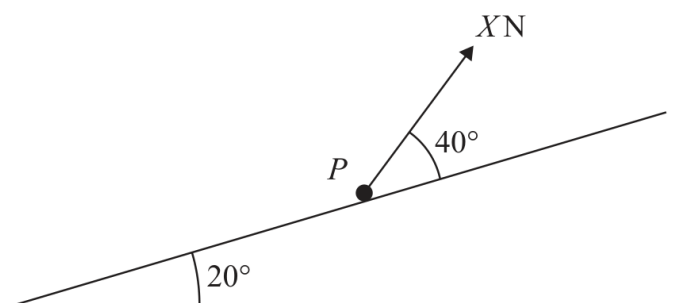


Figure 3

A particle P , whose weight is 50 N , is at rest on a rough plane. The plane is inclined to the horizontal at 20° and the coefficient of friction between P and the plane is 0.3

The particle P is held in equilibrium by a force of magnitude X newtons.

The line of action of the force makes an angle of 40° with the plane, as shown in Figure 3. The line of action of the force lies in a vertical plane which contains a line of greatest slope of the plane.

Given that P is on the point of sliding **down** the plane, find the value of X .

(6)

Summer 2026 Q5 — detailed solution

Oblique/horizontal force changes a rough-plane reaction

ALL
6 marks**Choose the limiting-friction direction**

Because P is about to slide down the plane, friction acts up the plane at its limiting value $0.3R$. The force has components $X \cos 40^\circ$ up the plane and $X \sin 40^\circ$ away from it.

Resolve perpendicular to the plane

Equilibrium normal to the plane gives $R + X \sin 40^\circ = 50 \cos 20^\circ$, so $R = 50 \cos 20^\circ - X \sin 40^\circ$.

Resolve parallel to the plane

Equilibrium along the plane gives $0.3R + X \cos 40^\circ = 50 \sin 20^\circ$. Hence $X(\cos 40^\circ - 0.3 \sin 40^\circ) = 50(\sin 20^\circ - 0.3 \cos 20^\circ)$.

Evaluate at the end

Solving gives $X = 5.243501 \dots \text{ N}$, so $X = 5.24 \text{ N}$ to three significant figures.

Final answer $X = 5.24 \text{ N}$.